Timing-Only Channel Estimation and Equalization for OFDM Through a Dispersive Time-Modulated RIS

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***Abstract*—A time-modulated reconfigurable intelligent surface (TM-RIS) switches its elements several times within each OFDM symbol. The switching couples every received subcarrier to many transmitted ones, and with frequency-dispersive elements the coupling matrix is neither diagonal nor Toeplitz. Existing TM-RIS designs treat this coupling as interference, which caps the legitimate user's signal-to-interference ratio at -4.8 dB for a 75% duty cycle, and the receiver for the dispersive model was left open. We show that grouping the harmonic expansion by element state separates the coupling exactly into Toeplitz factors fixed by the switching timing, which the receiver knows, and per-state input responses that absorb the propagation *and* the element dispersion, which it does not. From this separation we build, to our knowledge, the first pilot-based receiver for this channel: a low-rank estimator that needs no element model, a blind estimator of the RIS clock offset that rests on the fact that an offset is exactly an opposite delay of the two hops, and an exact matrix-free equalizer. In pre-registered tests with 30 channels per point, the receiver comes within 0.1 bit/s/Hz of perfect channel knowledge with two pilot symbols on a shared-path channel and needs four to eight on 3GPP TDL-C. At 20 and 30 dB it is within 0.02 bit/s/Hz of, or better than, every estimator that needs no training data on both channel families: element-model estimators with an ideal, a flat-calibrated or the exact element, a sparse estimator that uses the switching kernels, per-state static pilots, and basis-expansion estimators with up to all harmonics and 32 pilot symbols. Over a 7.7 MHz band two calibrated scalars per state make an element-model estimator as good as one given the exact element; ours matches it without calibration and is better at 30 dB (by 0.05 and 0.24 bit/s/Hz) because of its low rank. Over the full 98 MHz band, where the element is dispersive, the calibrated estimators have 1.5 times our symbol error rate at 28 dB, and one pilot symbol suffices. An LMMSE whose prior is learned from 300 channels of the test distribution is better than ours, but trained on a different delay spread it is worse by up to 0.95 bit/s/Hz; a prior learned from our own 8-pilot estimates, with no true channels and no element model, keeps most of the learned gain. Our receiver decodes coded blocks within 0.14 dB of perfect knowledge, and it absorbs physical clock offsets with no significant rate loss.**

***Index Terms*—Reconfigurable intelligent surface, time modulation, OFDM, intercarrier interference, channel estimation, equalization.**

I. Introduction

A time-modulated RIS (TM-RIS), or space-time-coding metasurface, switches each element several times within an OFDM symbol [1]. The reflected signal then carries harmonics of the switching rate, which has been used for waveform security and directional modulation [2], [3]. For the user the surface serves, the same harmonics move energy from transmitted subcarrier m to received subcarrier $m + h$, so the channel matrix is not diagonal. Iudice *et al.* [4] recently derived a physically consistent model of this channel with frequency-dispersive (finite-memory) elements, as real varactor cells are. They showed that the coupling survives any cyclic prefix, that with dispersive elements the coupling matrix is not even Toeplitz. They assume the channel known and name joint subcarrier equalization as future work [4, Sec. V]; channel estimation is not treated. Prior TM-RIS designs keep a one-tap receiver for the legitimate user and treat the coupling as interference [2], [3]. A blind eavesdropper attack [5] relies on the Toeplitz structure of ideal, frequency-flat switching, which dispersion breaks. Generic estimators of doubly selective OFDM channels, such as the basis-expansion model (CE-BEM) [6], do not know where the coupling comes from: they must estimate every harmonic. Because rectangular switching leaves a $1/h^2$ energy tail, they either truncate the harmonics, which caps their accuracy, or keep all of them, which needs many pilots.

Novelty. The receiver of a TM-RIS link knows *how* the channel varies inside the symbol: the variation is the switching pattern, which the network sets. It does not know the propagation or the exact element response. We turn this observation into an exact structural statement and build the whole receiver on it. Our contributions are:

1) *Timing–propagation separation* (Proposition 1). Grouping the expansion of [4, eq. (5)] by element state gives $\boldsymbol{H} = \sum_{q,s} \boldsymbol{T}_{q,s} \circ (\boldsymbol{a}_q^{\mathrm{d}} \boldsymbol{v}_{q,s}^{\mathsf{T}})$. The Toeplitz matrices $\boldsymbol{T}_{q,s}$ depend only on the switching timing. The element dispersion enters only the per-state input response $\boldsymbol{v}_{q,s}$, together with the unknown BS–RIS channel. The regrouping is elementary and we credit it to the anchor model; what is new is what it implies for a receiver.
2) *A receiver that needs no element model*: a low-rank, shared-subspace estimator of $\boldsymbol{H}$ that treats $\boldsymbol{v}_{q,s}$ as unknown (Algorithm 1). Unlike [5], it does not rely on Toeplitz structure; unlike CE-BEM, it never estimates harmonics separately.
3) *Blind clock-offset recovery.* We prove that an offset between the RIS and the receiver is exactly an opposite delay of the two hops (Proposition 2), and we identify

the offset physically (Proposition 3). This explains why a wrong offset overfits the pilots, and it leads to an estimator of the offset and the delay reference by Bayesian evidence (Algorithm 2).

4) *A pre-registered evaluation*: pass/fail criteria fixed before each run, 30 channels per point, paired confidence intervals, settings chosen on channels disjoint from the reported ones, the strongest model-based, structure-aware and learned baselines we could build (including a calibrated element model), ablations and a hyperparameter sensitivity study, and failures reported, including where a learned prior beats ours.

We adopt and do not claim: the DPSS delay basis [7], [8], low-rank RIS channel models [9], matrix-free iterative equalization [10] (exact here rather than a BEM approximation), and evidence-based model selection [11]. The closest timing work, blind offset estimation for RIS symbiotic radio [12], estimates the offset of a backscattered data stream from virtual subcarriers; our receiver instead uses the known pattern and needs no idle subcarriers.

Notation: $\circ$ and $\odot$ denote Hadamard products of matrices and vectors; $(\cdot)^{\mathsf{T}}$, $(\cdot)^{\mathsf{H}}$ and $(\cdot)^*$ denote transpose, conjugate transpose and conjugate; $\operatorname{sinc} x = \sin(\pi x)/(\pi x)$; $\mathcal{CN}$ denotes the circular complex Gaussian distribution.

II. System Model

A base station (BS) transmits OFDM with M subcarriers $f_m = f_c + m\Delta f$, $m \in \mathcal{M} = \{-M/2, \dots, M/2-1\}$, useful duration $T_u = 1/\Delta f$ and cyclic prefix (CP) T_{cp}, to a user through a RIS of Q elements (Fig. 1). Element q follows a periodic pattern $s_q[k] \in \{1,2\}$, $k = 0, \dots, K-1$, of period T_u: it is in state $s_q[k]$ during slot $[kT_u/K, (k+1)T_u/K)$. The duty cycle D is the fraction of slots in state 1. In state s the element reflects with the single-resonance response of [4],

$$\Gamma_s(f) = -1 + \frac{2\xi_{r,s}}{j2\pi(f - f_{0,s}) + \xi_{r,s} + \xi_{i,s}}, \tag{1}$$

which for the published OpenRIS cell ($f_{0,1} = 3.471\,\mathrm{GHz}$, $f_{0,2} = 3.710\,\mathrm{GHz}$, $\xi_{r,s}/2\pi = 97.5,\ 128.4\,\mathrm{MHz}$, $\xi_{i,s}/2\pi = 8.9,\ 9.0\,\mathrm{MHz}$ [4, Tab. I]) reproduces the in-channel behaviour reported there (Fig. 2): across the 100 MHz channel at $f_c = 3.594\,\mathrm{GHz}$ each state drifts by $46°$ and $49°$ in phase, the two states differ by $180° \pm 7.8°$ with up to 0.57 dB magnitude imbalance, and the reflection loss reaches 1.0 dB. The pair is thus a good binary phase shifter but not the frequency-flat, lossless ± 1 switch of earlier work. Over the 7.7 MHz band of our $M = 256$ comparisons the departure is mainly a static one (a phase difference of $172.2°$–$172.5°$); over the full band the dispersion matters as well. Let $a_q^{\mathrm{s}}(f)$ and $a_q^{\mathrm{d}}(f)$ be the BS–element and element–user responses. When the CP covers the cascaded delay spread plus the element memory (13.8 ns), the received frequency-domain symbol is [4]

$$\boldsymbol{y} = \boldsymbol{H}\boldsymbol{x} + \boldsymbol{w}, \quad H_{\bar{m}m} = \sum_{q=1}^{Q} a_q^{\mathrm{d}}(f_{\bar{m}})\, b_q^{[\bar{m}-m]}(f_m)\, a_q^{\mathrm{s}}(f_m), \tag{2}$$

TABLE I
Assumptions, justification, and where they are tested.

Assumption	Justification; test of its violation
A1: pattern $s_q[k]$ known	The network drives the RIS controller, so the pattern is control information, like a pilot configuration. Without it the coupling cannot be told apart from propagation: a receiver that ignores one of two panels collapses to 1.15 bit/s/Hz (Sec. V-G).
A2: generalized CP	Condition of [4]. Checked on the cascade: 3GPP TDL at 30 and 100 ns is valid (≤ -56 dB of energy beyond the CP); 300 ns is not and is out of scope.
A3: period T_u, aligned up to an offset d	TM-IRS designs synchronize the pattern to the symbol [2]; the residual offset is estimated (Sec. IV-C).
A4: shared paths	A compact surface sees the same paths; this is what makes the factors low-rank. Tested with 3GPP TDL taps from random directions, spatially consistent CDL, 1024 elements and a near-field user.
A5: channel constant over a refresh block	The coherence time $0.423/f_D$ [13] spans 1188, 119 and 30 symbols at 3, 30 and 120 km/h. Mobility is tested with the refresh interval selected under pilot overhead (Sec. V-F).
A6: σ^2 known; no mutual coupling	σ^2 is estimated from empty resource elements. Mutual coupling is ignored, as in [4]; this is a stated limitation.

where $\boldsymbol{w} \sim \mathcal{CN}(\boldsymbol{0}, \sigma^2\boldsymbol{I})$ and $b_q^{[h]}(f)$ is the h-th Fourier coefficient, over one period, of the switched response of element q at input frequency f. Equation (2) is the ground truth of our evaluation. Its implementation agrees with an independent time-domain simulation (filter, gate, FFT, with no harmonic algebra) to 0.27% median relative error.

Assumptions. Table I lists each assumption with its justification and where its violation is evaluated. The receiver does *not* assume (1): it only defines the truth.

III. Structure of the Coupling

Proposition 1 (Timing–propagation separation)**.** *For a K-slot pattern of period T_u, $b_q^{[h]}(f) = \sum_{s=1}^{2} t_{q,s}[h]\Gamma_s(f)$ with*

$$t_{q,s}[h] = \frac{e^{-j\pi h/K}}{K} \operatorname{sinc}\left(\frac{h}{K}\right) \sum_{k:\, s_q[k]=s} e^{-j2\pi hk/K}. \tag{3}$$

Hence, with $[\boldsymbol{T}_{q,s}]_{\bar{m}m} = t_{q,s}[\bar{m}-m]$ (Toeplitz), $\boldsymbol{a}_q^{\mathrm{d}} = [a_q^{\mathrm{d}}(f_m)]_m$ and $\boldsymbol{v}_{q,s} = \boldsymbol{\Gamma}_s \odot \boldsymbol{a}_q^{\mathrm{s}}$,

$$\boldsymbol{H} = \sum_{q=1}^{Q}\sum_{s=1}^{2} \boldsymbol{T}_{q,s} \circ \left(\boldsymbol{a}_q^{\mathrm{d}} \boldsymbol{v}_{q,s}^{\mathsf{T}}\right). \tag{4}$$

Proof. In slot k the element applies $\Gamma_{s_q[k]}(f)$, so its switched response is $\sum_s \Gamma_s(f)\mathbb{1}_s(t)$, where $\mathbb{1}_s$ indicates the slots in state s. The h-th Fourier coefficient of the indicator of slot k is $\frac{1}{T_u}\int_{kT_u/K}^{(k+1)T_u/K} e^{-j2\pi ht/T_u}dt = \frac{1}{K}e^{-j\pi h/K}\operatorname{sinc}(h/K)e^{-j2\pi hk/K}$. Summing over the slots in state s gives (3). Substituting into (2), $t_{q,s}$ depends on $\bar{m}-m$ only, which gives (4). □

Fig. 3 shows the resulting structure: the main diagonal is the zero-order reflection and the off-diagonals are the harmonics,

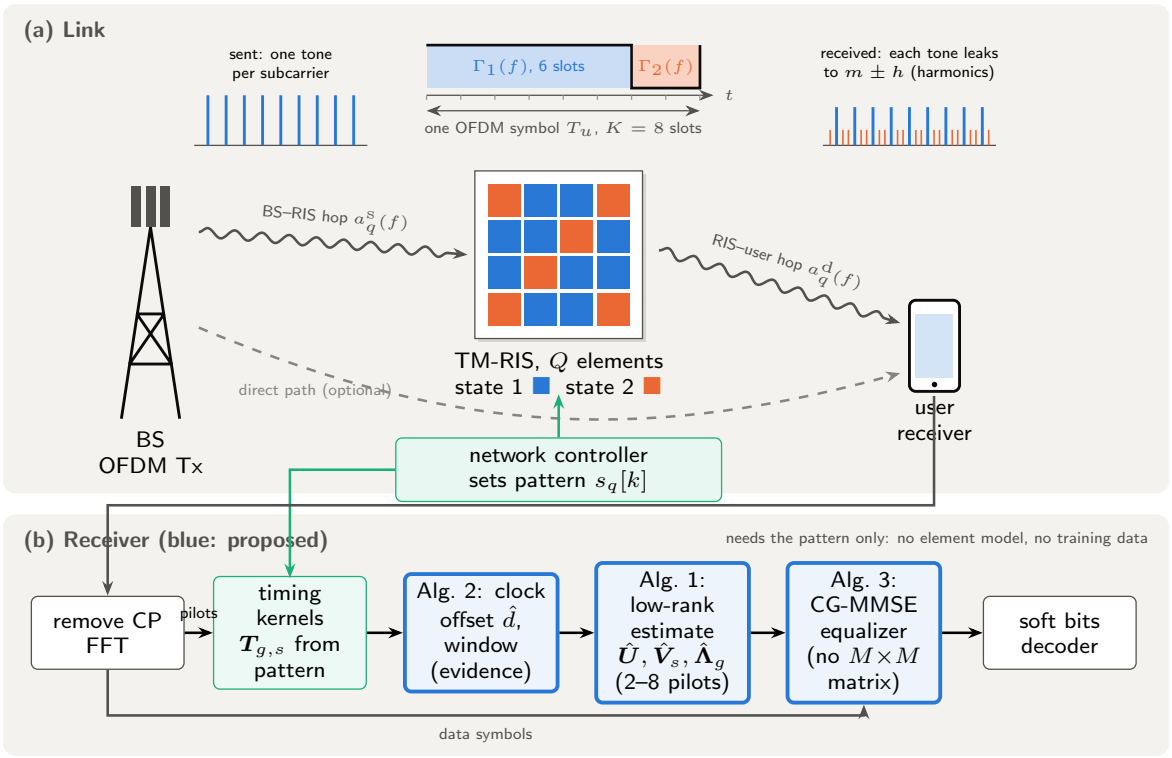


Fig. 1. (a) The link: each RIS element switches between two states within every OFDM symbol (here $K = 8$ slots, duty cycle $D = 0.75$), so every transmitted subcarrier reaches the user spread over harmonic offsets. (b) The receiver: it uses the switching pattern from the network to build the timing kernels, never the element model, and it never forms the $M \times M$ coupling matrix. Blue: the proposed algorithms.

whose envelope follows $|\operatorname{sinc}(h/K)|$; for $K = 2$ the even harmonics vanish, which our implementation reproduces from [4, Fig. 5].

Proposition 1 has five consequences, which together define the receiver.

Corollary 1 (No element model)**.** *The element response enters* (4) *only through $\boldsymbol{v}_{q,s}$, which also contains the unknown BS–RIS channel. Estimating $\boldsymbol{v}_{q,s}$ as an unknown absorbs the dispersion exactly; the receiver needs only the timing kernels $\boldsymbol{T}_{q,s}$.*

Corollary 2 (One-tap ceiling)**.** *For ideal ± 1 elements with a common pattern of duty D, $b^{[0]} = t_1[0] - t_2[0] = 2D - 1$, and Parseval's theorem gives $\sum_h |b^{[h]}|^2 = \frac{1}{T_u}\int_0^{T_u} |\pm 1|^2 dt = 1$. A one-tap receiver collects the fraction $(2D-1)^2$ of the reflected energy and sees the rest as interference, so its SIR is at most $(2D-1)^2/(1-(2D-1)^2)$: $-4.8\,dB$ at $D = 0.75$. Joint equalization collects all of it.*

Corollary 3 (Harmonic tail)**.** *From* (3)*, $|t_{q,s}[h]| \le |\operatorname{sinc}(h/K)|$, which decays as $1/|h|$, so the energy outside $|h| \le H$ decays as $1/H$. For $K = 8$ and $D = 0.75$ it is 3.4%, 1.0% and 0.32% for $H = 12$, 40 and 128 (Fig. 3c). A model truncated to $|h| \le 12$ therefore has a normalized error floor of $10\log_{10} 0.034 = -14.7\,dB$, whatever its estimator.*

Corollary 4 (Fast products)**.** *$(\boldsymbol{T} \circ (\boldsymbol{a}\boldsymbol{b}^{\mathsf{T}}))\boldsymbol{x} = \boldsymbol{a} \odot (\boldsymbol{T}(\boldsymbol{b} \odot \boldsymbol{x}))$, and $\boldsymbol{T}\boldsymbol{z}$ is a linear convolution. Hence $\boldsymbol{H}\boldsymbol{x}$ and $\boldsymbol{H}^{\mathsf{H}}\boldsymbol{y}$ cost $\mathcal{O}(QM \log M)$ with FFTs of length at least $3M - 2$, and $\boldsymbol{H}$ is never formed.*

Proposition 2 (An offset is an opposite delay)**.** *If the switching runs d slots late relative to the receiver's timing, each kernel becomes $\boldsymbol{D}\boldsymbol{T}\boldsymbol{D}^*$ with $\boldsymbol{D} = \operatorname{diag}(e^{-j2\pi md/K})_{m \in \mathcal{M}}$, and*

$$(\boldsymbol{D}\boldsymbol{T}\boldsymbol{D}^*) \circ (\boldsymbol{a}\boldsymbol{b}^{\mathsf{T}}) = \boldsymbol{T} \circ \big((\boldsymbol{D}\boldsymbol{a})(\boldsymbol{D}^*\boldsymbol{b})^{\mathsf{T}}\big). \tag{5}$$

Multiplying a frequency response by $\boldsymbol{D}$ delays it by dT_u/K; multiplying it by $\boldsymbol{D}^*$ advances it by the same amount.

Proof. A shift of d slots multiplies the h-th coefficient by $e^{-j2\pi hd/K} = e^{-j2\pi \bar{m}d/K} e^{j2\pi md/K}$ for $h = \bar{m} - m$. This is (5) entrywise. □

Proposition 3 (Physical timing)**.** *Let the two hops have bulk delays τ_b and τ_u, and let the receiver synchronize to the first cascaded arrival. (i) If the RIS controller is locked to the signal arriving at the RIS, the synchronized $\boldsymbol{H}$ equals the one without bulk delays. (ii) If it is locked to the BS clock, the synchronized $\boldsymbol{H}$ equals the one without bulk delays and with offset $d = -\tau_b K/T_u$.*

Proof. The delays multiply the input side by $e^{-j2\pi f_m \tau_b}$ and the output side by $e^{-j2\pi f_{\bar{m}} \tau_u}$; synchronization multiplies the output by $e^{j2\pi f_{\bar{m}}(\tau_b + \tau_u)}$. In (i) the switching edges are delayed by τ_b, which by Proposition 2 contributes $e^{-j2\pi(\bar{m}-m)\Delta f \tau_b}$, and all phases cancel. In (ii) the edges are not delayed, and what remains is the phase of an offset of $-\tau_b$. □

Both identities hold numerically to $7 \cdot 10^{-15}$. A practical RIS therefore sees either no offset or an offset equal to its BS distance, tens to hundreds of nanoseconds; the latter costs 1.6/4.2 bit/s/Hz at 20/30 dB if ignored (Table V).

IV. The Receiver

A. Low-rank model with a delay prior

Under A4, the user-hop response of element q is $a_q^{\mathrm{d}}(f) = \sum_\ell g_\ell e^{-j\boldsymbol{k}\boldsymbol{u}_\ell^{\mathsf{T}}\boldsymbol{r}_q} e^{-j2\pi f\tau_\ell}$, so $\boldsymbol{a}_q^{\mathrm{d}} = \boldsymbol{E}\boldsymbol{\alpha}_q$ with $\boldsymbol{E} = [e^{-j2\pi f_m\tau_\ell}]_{m,\ell}$ common to all elements. Likewise $\boldsymbol{v}_{q,s} = \mathrm{diag}(\boldsymbol{\Gamma}_s)\boldsymbol{E}'\boldsymbol{\beta}_q$, where $\boldsymbol{\beta}_q$ (the input-hop path gains with the element's geometric phases) is common to both states. Hence $[\boldsymbol{a}_1^{\mathrm{d}}, \dots, \boldsymbol{a}_Q^{\mathrm{d}}]$ has rank at most the number of paths, whatever Q. Every hop response is delay-limited to a window $[t_0, t_0 + \tau_{\max}]$, where $\tau_{\max}$ also covers the element memory. We represent the responses with the first L discrete prolate spheroidal sequences (DPSS) [7], [8]. These are the columns of $\boldsymbol{F} \in \mathbb{C}^{M\times L}$: the leading eigenvectors of $R_{nm} = e^{-j\pi(n-m)\Delta f\tau_{\max}}\,\mathrm{sinc}((n-m)\Delta f\tau_{\max})$, the frequency correlation of a response with delays uniform in the window. Their normalized eigenvalues $\boldsymbol{\lambda}$ serve as prior variances. Compressing the path gains to rank r gives $\boldsymbol{a}_q^{\mathrm{d}} = \boldsymbol{F}\boldsymbol{U}\tilde{\boldsymbol{\alpha}}_q$ and $\boldsymbol{v}_{q,s} = \boldsymbol{F}\boldsymbol{V}_s\tilde{\boldsymbol{\beta}}_q$. The second relation holds because Γ_s is smooth (memory 13.8 ns): $\mathrm{diag}(\boldsymbol{\Gamma}_s)\boldsymbol{E}'$ stays delay-limited, so it lies in the same DPSS span with a state-specific coefficient matrix $\boldsymbol{V}_s$, while $\tilde{\boldsymbol{\beta}}_q$ is shared by the two states. With G distinct patterns (groups $\mathcal{G}_g$) and rank r, (4) becomes

$$\boldsymbol{H} = \sum_{g=1}^{G}\sum_{s=1}^{2} \boldsymbol{T}_{g,s} \circ \big((\boldsymbol{F}\boldsymbol{U})\boldsymbol{\Lambda}_g(\boldsymbol{F}\boldsymbol{V}_s)^{\mathsf{T}}\big), \tag{6}$$

with $\boldsymbol{U}, \boldsymbol{V}_s \in \mathbb{C}^{L\times r}$ and $\boldsymbol{\Lambda}_g = \sum_{q\in\mathcal{G}_g} \tilde{\boldsymbol{\alpha}}_q\tilde{\boldsymbol{\beta}}_q^{\mathsf{T}} \in \mathbb{C}^{r\times r}$. The number of unknowns falls from up to M^2 to $3Lr + Gr^2$. An unswitched direct path is the special case of equal input responses in both states (since $\boldsymbol{T}_{q,1}+\boldsymbol{T}_{q,2} = \boldsymbol{I}$), so (6) contains it without change.

B. Timing-only estimation (Algorithm 1)

Let pilot symbol p be $\boldsymbol{s}_p$, with received $\boldsymbol{y}_p$, and let $\boldsymbol{G}_{g,s,p} = \boldsymbol{T}_{g,s}\,\mathrm{diag}(\boldsymbol{s}_p)\boldsymbol{F}$, computed once by FFT. By Corollary 4,

$$\boldsymbol{H}\boldsymbol{s}_p = \sum_{g,s}\big[(\boldsymbol{F}\boldsymbol{U}\boldsymbol{\Lambda}_g) \circ (\boldsymbol{G}_{g,s,p}\boldsymbol{V}_s)\big]\mathbf{1}_r, \tag{7}$$

which is linear in $\boldsymbol{U}$, in $[\boldsymbol{V}_1; \boldsymbol{V}_2]$ and in $\{\boldsymbol{\Lambda}_g\}$ when the other two are fixed. For each block $\boldsymbol{\theta}$ the N_p stacked pilots give $\boldsymbol{y} = \boldsymbol{A}_{\boldsymbol{\theta}}\boldsymbol{\theta} + \boldsymbol{w}$. With the prior $\boldsymbol{\theta} \sim \mathcal{CN}(\mathbf{0}, \boldsymbol{P})$, $\boldsymbol{P} = \mathrm{diag}(\boldsymbol{p})$, the columns of $\boldsymbol{U}$ and $\boldsymbol{V}_s$ carry variances $\varsigma\boldsymbol{\lambda}$, with the scale ς set by moment matching on $\|\boldsymbol{y}\|^2$. The LMMSE update

$$\hat{\boldsymbol{\theta}} = \boldsymbol{P}^{1/2}\big(\boldsymbol{P}^{1/2}\boldsymbol{A}_{\boldsymbol{\theta}}^{\mathsf{H}}\boldsymbol{A}_{\boldsymbol{\theta}}\boldsymbol{P}^{1/2} + \sigma^2\boldsymbol{I}\big)^{-1}\boldsymbol{P}^{1/2}\boldsymbol{A}_{\boldsymbol{\theta}}^{\mathsf{H}}\boldsymbol{y} \tag{8}$$

stays well posed when some prior variances are tiny. Alternating (8) over the blocks is non-convex, so the initialization matters. Ignoring the coupling between groups gives a *linear* model, $\boldsymbol{H} \approx \sum_s \bar{\boldsymbol{T}}_s \circ (\boldsymbol{F}\boldsymbol{C}_s\boldsymbol{F}^{\mathsf{T}})$ with $\bar{\boldsymbol{T}}_s = \sum_g \boldsymbol{T}_{g,s}$, which is exact for $G = 1$. Its LMMSE estimate with prior $\varsigma(\boldsymbol{\lambda}\otimes\boldsymbol{\lambda})$, followed by a rank-$r$ SVD $[\hat{\boldsymbol{C}}_1\ \hat{\boldsymbol{C}}_2] \approx \boldsymbol{Y}\boldsymbol{\Sigma}\boldsymbol{W}^{\mathsf{H}}$, initializes the factors. From this initialization ALS converges in one or two iterations; random initializations end 3–5 times worse.

Algorithm 1 Timing-only low-rank estimation

Input: pilots $\{\boldsymbol{y}_p, \boldsymbol{s}_p\}$; kernels $\boldsymbol{T}_{g,s}$; $\boldsymbol{F}, \boldsymbol{\lambda}$; σ^2; rank r; iterations I

1: $\boldsymbol{G}_{g,s,p} \leftarrow \boldsymbol{T}_{g,s}\,\mathrm{diag}(\boldsymbol{s}_p)\boldsymbol{F}$ by FFT
2: $[\hat{\boldsymbol{C}}_1, \hat{\boldsymbol{C}}_2] \leftarrow$ LMMSE of the linear model; rank-r SVD $\rightarrow \boldsymbol{U} = \boldsymbol{Y}_r\boldsymbol{\Sigma}_r^{1/2}$, $[\boldsymbol{V}_1; \boldsymbol{V}_2] = \boldsymbol{W}_r^*\boldsymbol{\Sigma}_r^{1/2}$, $\boldsymbol{\Lambda}_g = \boldsymbol{I}$
3: **for** $i = 1, \dots, I$ **do**
4: $\quad\{\boldsymbol{\Lambda}_g\} \leftarrow$ ridge least squares on (7); $\boldsymbol{U} \leftarrow$ (8); $[\boldsymbol{V}_1; \boldsymbol{V}_2] \leftarrow$ (8)

Output: factors of (6) ($\boldsymbol{H}$ is not formed)

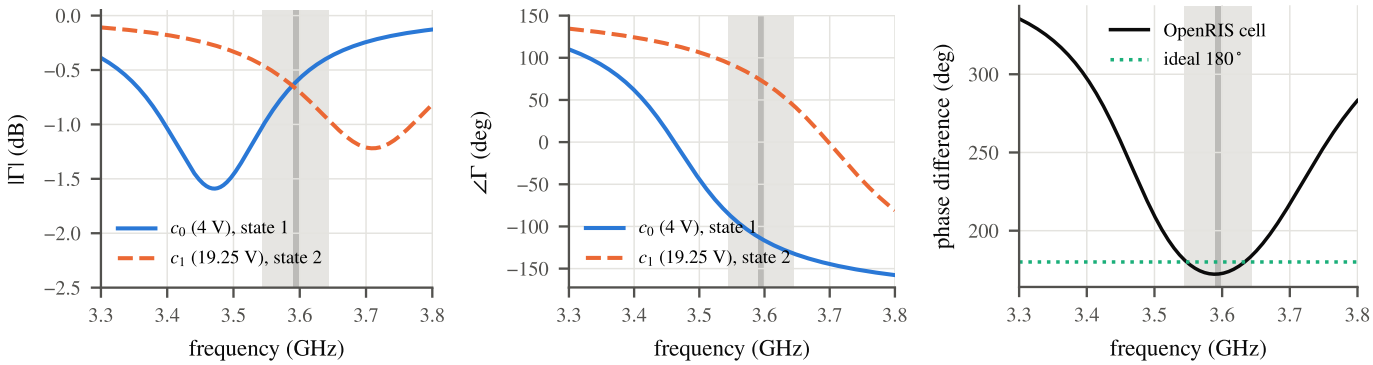
Why two pilot symbols can suffice. At $M = 256$ with $L = 16$ and $r = 8$ there are 448 unknowns against 512 observations from two pilot symbols, yet the problem is well determined because the prior concentrates each hop on about $M\Delta f\tau_{\max}+1$ dimensions. We measured the effective degrees of freedom of the linear stage, the trace of its LMMSE hat matrix: 19–27 at 20 dB and 23–33 at 30 dB for 1–8 pilot symbols, against 256–2048 observations. With two pilot symbols the achieved coupling-matrix error is 5–6 dB above the Bayesian bound under the assumed prior (-26.4 vs -32.4 dB at 20 dB). The gap is not estimator suboptimality: real channels do not follow the uniform-delay prior. On TDL-C ($L = 18$, $r = 16$) the low-rank model has 1120 unknowns against 1024 observations from four pilot symbols, and the effective degrees of freedom of the linear stage are 58, 81 and 102 at 10, 20 and 30 dB. The two-pilot figure is therefore a property of the shared-path family, which is nearly flat over a 7.7 MHz band, not of the method; at the full band one pilot symbol suffices on it (Sec. V-E). An earlier version kept the linear estimate whenever it fitted the pilots better. On fresh 3GPP channels this rule was an overfitting trap (chosen in 37–57% of channels at 30 dB, worse rate), so Algorithm 1 returns the low-rank estimate. The week-10 to week-12 runs behind Figs. 4–8 and Tables III and VI still applied the rule; on 30 fresh channels per family it was never triggered on the shared path and changed the TDL-C loss by $+0.004 \pm 0.005$ / $+0.019 \pm 0.024$ at 20/30 dB, so those results are those of Algorithm 1 within their intervals.

C. Blind clock offset and delay reference (Algorithm 2)

By Proposition 2 a candidate offset d replaces each kernel by $\boldsymbol{D}_d\boldsymbol{T}_{g,s}\boldsymbol{D}_d^*$, and a window start t_0 replaces $\boldsymbol{F}$ by $\boldsymbol{F}_{t_0} = \mathrm{diag}(e^{-j2\pi m\Delta f t_0})\boldsymbol{F}$. For each candidate the linear stage is a Gaussian model, $\boldsymbol{y} = \boldsymbol{\Phi}\boldsymbol{c} + \boldsymbol{w}$ with $\boldsymbol{c} \sim \mathcal{CN}(\mathbf{0}, \boldsymbol{P})$, and its log-evidence is, up to a constant,

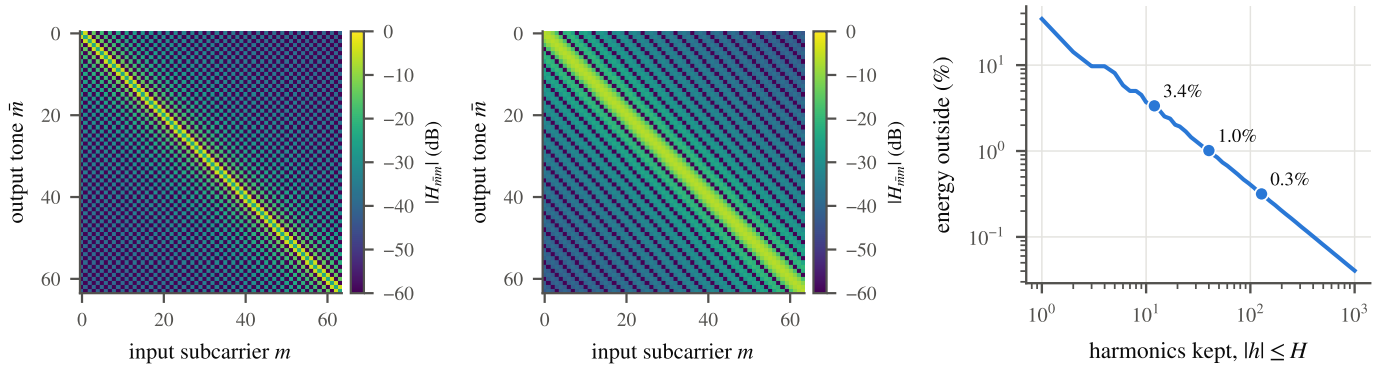
$$\mathcal{E}(d, t_0) = -\boldsymbol{y}^{\mathsf{H}}\boldsymbol{C}^{-1}\boldsymbol{y} - \log\det\boldsymbol{C}, \qquad \boldsymbol{C} = \boldsymbol{\Phi}\boldsymbol{P}\boldsymbol{\Phi}^{\mathsf{H}} + \sigma^2\boldsymbol{I}. \tag{9}$$

With the Cholesky factor $\boldsymbol{R}^{\mathsf{H}}\boldsymbol{R} = \sigma^{-2}\boldsymbol{P}^{1/2}\boldsymbol{\Phi}^{\mathsf{H}}\boldsymbol{\Phi}\boldsymbol{P}^{1/2} + \boldsymbol{I}$ and the Woodbury identity, $\log\det\boldsymbol{C} = MN_p\log\sigma^2 + 2\sum_i \log R_{ii}$ and $\boldsymbol{y}^{\mathsf{H}}\boldsymbol{C}^{-1}\boldsymbol{y} = \sigma^{-2}\|\boldsymbol{y}\|^2 -$



(a) reflection magnitude (b) reflection phase (c) phase difference of the states

Fig. 2. The OpenRIS cell in the format of [4, Figs. 2–4]: single-pole model (1) with the parameters and common phase $-10.7°$ of [4, Tab. I]. Light band: the 100 MHz channel; dark strip: the 7.7 MHz band of the $M = 256$ comparisons.



(a) $K = 2$ (setting of [4, Fig. 5]) (b) $K = 8$, $D = 0.75$ (this paper) (c) harmonic tail, $K = 8$, $D = 0.75$

Fig. 3. Coupling-matrix magnitude for one element with ideal single-tap channels, $M = 64$, OpenRIS cell: (a) our implementation of (2) for the setting of [4, Fig. 5], reproducing it; (b) the setting of this paper. (c) Fraction of the coupling energy outside $|h| \le H$ (Corollary 3).

$\sigma^{-4}\|\boldsymbol{R}^{-\mathsf{H}}\boldsymbol{P}^{1/2}\boldsymbol{\Phi}^{\mathsf{H}}\boldsymbol{y}\|^2$, so the evidence costs one $2L^2$-dimensional factorization per candidate.

Why the evidence. By (5), a wrong offset can still explain the pilots, through a delayed user hop and an advanced BS hop. The pilot residual is therefore nearly flat in d for physical offsets below about 100 ns. For such offsets, the residual of the advanced response after projection onto the delay basis is only $2.7 \cdot 10^{-4}$ at 100 ns (relative). A wrong offset thus *overfits*: it fits through improbable shifted responses. The log-determinant in (9) penalizes exactly this. With two held-out pilot symbols, the offset that minimizes the held-out residual agrees with the evidence maximizer within 0.005 slot in 93% of channels. The window start must be estimated too. After synchronization the first arrivals of the two hops sum to zero up to the offset, so a window that starts after the true reference cannot hold both hops; a synchronization error has the same effect. The evidence of different priors is directly comparable, so t_0 is chosen jointly with d.

Algorithm 2 Offset and delay reference by evidence

Input: pilots; nominal kernels; $\boldsymbol{F}, \boldsymbol{\lambda}$; σ^2; $\mathcal{T}_0 = \{-150, -75, 0, 75\}$ ns

1: **for** $t_0 \in \mathcal{T}_0$ **do**

2: $d_0 \leftarrow \arg\max_d \mathcal{E}(d, t_0)$ on a 41-point grid over $[-\frac{1}{2}, \frac{1}{2}]$ slot

3: $\hat{d}_{t_0} \leftarrow$ golden-section search of $\mathcal{E}(\cdot, t_0)$ on $d_0 \pm 0.025$ (14 steps)

Output: $(\hat{d}, \hat{t}_0) = \arg\max_{t_0} \mathcal{E}(\hat{d}_{t_0}, t_0)$; run Algorithm 1 with them

D. Matrix-free equalization and complexity

The MMSE symbol estimate solves $(\hat{\boldsymbol{H}}^{\mathsf{H}}\hat{\boldsymbol{H}} + \sigma^2\boldsymbol{I})\boldsymbol{x} = \hat{\boldsymbol{H}}^{\mathsf{H}}\boldsymbol{y}$. We solve it by conjugate gradients (CG), as in [10]. Each product goes through (6): for $G = 1$, the FFTs of all r columns and all data symbols form one batched call per state, the two states are summed in the frequency domain, and one batched inverse FFT follows (Algorithm 3). Unlike the BEM approximation in [10], this representation is exact. The stopping tolerance was chosen by a pre-registered sweep. With $\epsilon = 10^{-2}$ the symbol error rate (SER) is 1.15 times the dense solve's at $M = 3276$; with $\epsilon = 3 \cdot 10^{-3}$ it is within 2% of it at every M from 1024 to 6552, after 8–20 iterations.

Complexity (complex multiply–adds). One ALS iteration costs $\sum_{n \in \{Lr, 2Lr, Gr^2\}} (MN_p n^2 + n^3/3)$; the linear stage costs $MN_p(2L^2)^2 + (2L^2)^3/3$. For $G = 1$, $r = 8$, $N_p = 2$ and 20 iterations this totals $5.7 \cdot 10^8$ at $M = 256$ and

Algorithm 3 Matrix-free CG-MMSE equalization

Input: data $\boldsymbol{Y}_d$ (symbols as columns); factors of $\hat{\boldsymbol{H}}$; σ^2; $\epsilon = 3 \cdot 10^{-3}$; $i_{\max} = 50$

$\boldsymbol{B} \leftarrow \hat{\boldsymbol{H}}^{\mathsf{H}} \boldsymbol{Y}_d$; $\boldsymbol{X} \leftarrow \boldsymbol{0}$; $\boldsymbol{R} \leftarrow \boldsymbol{B}$; $\boldsymbol{P} \leftarrow \boldsymbol{R}$

for $i = 1, \ldots, i_{\max}$ **do** ▷ per column

  $\boldsymbol{Q} \leftarrow \hat{\boldsymbol{H}}^{\mathsf{H}}(\hat{\boldsymbol{H}}\boldsymbol{P}) + \sigma^2 \boldsymbol{P}$ ▷ batched FFT products

  $\alpha \leftarrow \|\boldsymbol{r}\|^2 / \boldsymbol{p}^{\mathsf{H}}\boldsymbol{q}$; $\boldsymbol{x} \leftarrow \boldsymbol{x} + \alpha \boldsymbol{p}$; $\boldsymbol{r}' \leftarrow \boldsymbol{r} - \alpha \boldsymbol{q}$

  if $\|\boldsymbol{r}'\|/\|\boldsymbol{b}\| < \epsilon$ for all columns **break**

  $\boldsymbol{p} \leftarrow \boldsymbol{r}' + (\|\boldsymbol{r}'\|^2/\|\boldsymbol{r}\|^2)\boldsymbol{p}$; $\boldsymbol{r} \leftarrow \boldsymbol{r}'$

$3.9 \cdot 10^{10}$ at the full $M = 3276$. One CG iteration costs two products of $\mathcal{O}(rM \log M)$, together about $1.7 \cdot 10^7$ operations at $M = 3276$, so a symbol costs at most $3.4 \cdot 10^8$ with 20 iterations. The dense alternative forms $\hat{\boldsymbol{H}}$ (172 MB at $M = 3276$, 687 MB at $M = 6552$) and factors $\hat{\boldsymbol{H}}^{\mathsf{H}}\hat{\boldsymbol{H}} + \sigma^2 \boldsymbol{I}$ once per estimate ($4.7 \cdot 10^{10}$ operations), after which each symbol costs $\mathcal{O}(M^2)$. Measured on an idle 4-thread CPU ($\epsilon = 3 \cdot 10^{-3}$, $M = 3276$), dense equalization of 1, 12, 40 and 120 symbols per estimate took 2.51, 2.54, 2.58 and 2.78 s and CG 0.23, 2.51, 8.2 and 23.8 s, with CG's SER within 2% of dense. The break-even is thus about 12 symbols, not the 140 the operation count suggests, because the dense factorization runs as level-3 BLAS. CG is the faster choice for the 6–15-symbol blocks selected at 30 and 120 km/h (Sec. V-F); dense is 3–9× faster for slow channels. CG's memory is linear in M: at $M = 6552$ it equalized 120 symbols in 51 s without the 687 MB matrix. Warm-starting a refresh from the previous factors (two ALS iterations, no linear stage) took 1.4 s against 12.5 s from scratch, with the same SER.

V. Evaluation

Setup. We use the n78 carrier with $\Delta f = 30$ kHz, $T_{\mathrm{cp}} = 2.34\,\mu$s, $Q = 16$ elements (4×4, $\lambda/2$), $K = 8$, $D = 0.75$, identical patterns unless stated, and the OpenRIS cell (1). Comparisons use $M = 256$ (central 128 subcarriers scored); full-band results use $M = 3276$. Channels:

- a *shared-path* channel: per hop, one line-of-sight and five scattered paths, 30 ns rms, Rician factor 10 dB;
- 3GPP TDL-A/C/D [14] at 100 ns, with taps from random directions.

Pilots are QPSK pilot symbols. The estimator uses $\tau_{\max} = 0.3\,\mu$s, $L = 16$, $r = 8$ on the shared-path channel and $\tau_{\max} = 1\,\mu$s, $L = \lceil M\Delta f \tau_{\max} \rceil + 10$, $r = 16$ on TDL. These hyperparameters were frozen before the tests below. The metric is the achievable rate of the MMSE filter $\boldsymbol{W}$ built from the estimate and applied to the *true* channel: with $\boldsymbol{G} = \boldsymbol{W}\boldsymbol{H}$, $\mathrm{SINR}_m = |G_{mm}|^2/(\sum_{n \neq m} |G_{mn}|^2 + \sigma^2 \|\boldsymbol{w}_m\|^2)$, and $R = \mathrm{mean}_m \log_2(1 + \mathrm{SINR}_m)$. The *loss* is R with perfect channel state information (CSI) minus R with the estimate. Each test was pre-registered with its pass criterion and uses 30 independent channels per point; intervals are paired 95% t-intervals.

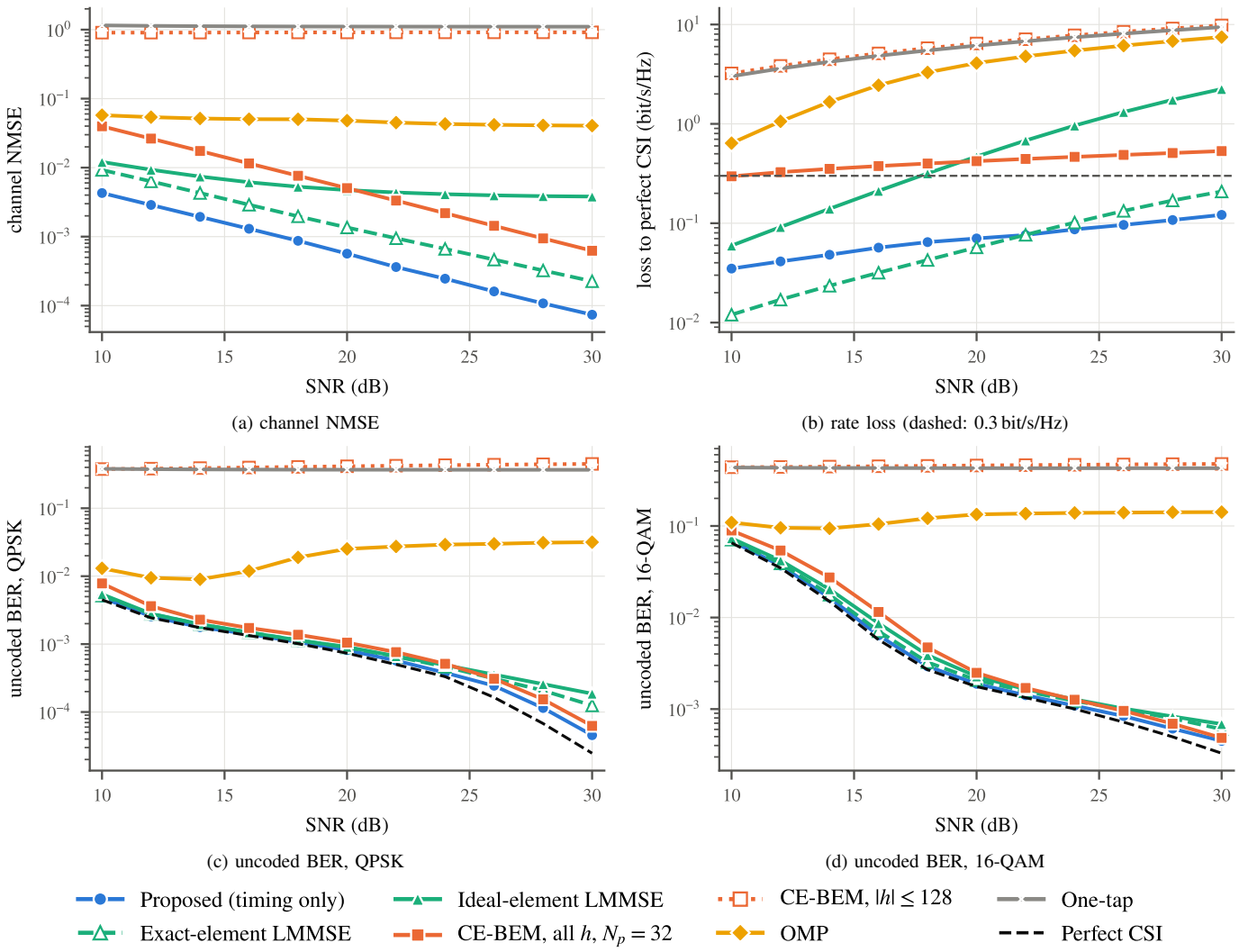
Baselines. All baselines get the same pilots, the same prior family and, for fairness, three equalizers each: naive MMSE, robust MMSE with the estimation-error power taken from the pilot residual, and robust MMSE with the oracle error power. Figures show the residual-based robust MMSE, which is usable in practice; paired comparisons take, per channel, the best over all baselines and all three equalizers.

- *One-tap*: today's OFDM receiver.
- *Ideal-element LMMSE*: the strongest estimator that assumes the textbook element, $\boldsymbol{H} = (\boldsymbol{T}_1 - \boldsymbol{T}_2) \circ (\boldsymbol{F}\boldsymbol{C}\boldsymbol{F}^{\mathsf{T}})$ with our prior. It tests the ± 1 assumption.
- *Flat-calibrated element LMMSE*: the same with per-state complex scalars, $\boldsymbol{H} = (g_1\boldsymbol{T}_1 + g_2\boldsymbol{T}_2) \circ (\boldsymbol{F}\boldsymbol{C}\boldsymbol{F}^{\mathsf{T}})$, $g_s = \Gamma_s(f_c)$ (calibrated), or with g_2/g_1 estimated from the pilots through our element-free linear stage (*flat-estimated*).
- *Structure-aware baselines*: OMP over $\boldsymbol{T}_s \circ (\boldsymbol{e}(\tau_a)\boldsymbol{e}(\tau_b)^{\mathsf{T}})$ on a 32.6 ns delay grid (it uses the known kernels); and *per-state static pilots*, where the RIS holds each state during half of the pilot symbols, the diagonal channel $\boldsymbol{h}_s$ is estimated by LMMSE, and Proposition 1 is applied with the channel assumed flat over the harmonic span, $\boldsymbol{H} \approx \sum_s \boldsymbol{T}_s \,\mathrm{diag}\, \boldsymbol{h}_s$.
- *CE-BEM* [6]: $\boldsymbol{H} = \sum_{|h| \le H_b} \boldsymbol{S}_h \,\mathrm{diag}(\boldsymbol{F}_2 \boldsymbol{c}_h)$ with up to all $2M - 1$ harmonics, solved matrix-free (equal to the dense solver to $7 \cdot 10^{-9}$); with all harmonics it is run from eight pilot symbols.
- *OMP* [15] over a harmonic×delay dictionary.
- *Learned-prior LMMSE*, the learning-based baseline: a principal-component basis of $\mathrm{vec}(\boldsymbol{H})$ and its eigenvalues learned from N_{tr} true channels form a Gaussian prior, and the LMMSE estimate follows from the pilots. It is the Bayes-optimal linear estimator for the learned statistics, which a network trained for minimum mean squared error approaches for near-Gaussian channels, and it learns the element non-ideality from data (Sec. V-B).

Three references are not practical receivers but locate the methods between the extremes: the *exact-element LMMSE*, which is the ideal-element LMMSE given the true responses, $\boldsymbol{H} = (\boldsymbol{T}_1 \,\mathrm{diag}\, \boldsymbol{\Gamma}_1 + \boldsymbol{T}_2 \,\mathrm{diag}\, \boldsymbol{\Gamma}_2) \circ (\boldsymbol{F}\boldsymbol{C}\boldsymbol{F}^{\mathsf{T}})$; the same estimator with both resonances shifted by 5–20 MHz (a mismatched model); and the *exact element with low rank*, Algorithm 1 given the true responses, the honest upper reference for ours.

A. Comparison with the strongest baselines

At low SNR the receivers that know the element model are slightly better than ours (by ≤ 0.03 bit/s/Hz on the shared path, ≤ 0.06 on TDL-C), since a linear LMMSE with the right structure needs no nonlinear fit. The order reverses between 17 and 22 dB, depending on the channel set (22 dB for the shared-path channels of Fig. 4, 17 dB for TDL-C), and the gaps then grow. On the channels of the original 0–30 dB comparison, at 20/30 dB ours loses 0.08/0.17 bit/s/Hz, the exact-element LMMSE 0.10/0.36, the 20 MHz-mismatched model 0.13/0.56, CE-BEM with 16 times our pilots 0.44/0.57, the ideal-element LMMSE 0.50/2.32 and OMP 4.1/7.5. Ours beats the exact-element LMMSE by having fewer unknowns (rank $r = 8$ against $L^2 = 256$), not by knowing more. On TDL-C ($N_p =$



(a) channel NMSE

(b) rate loss (dashed: 0.3 bit/s/Hz)

(c) uncoded BER, QPSK

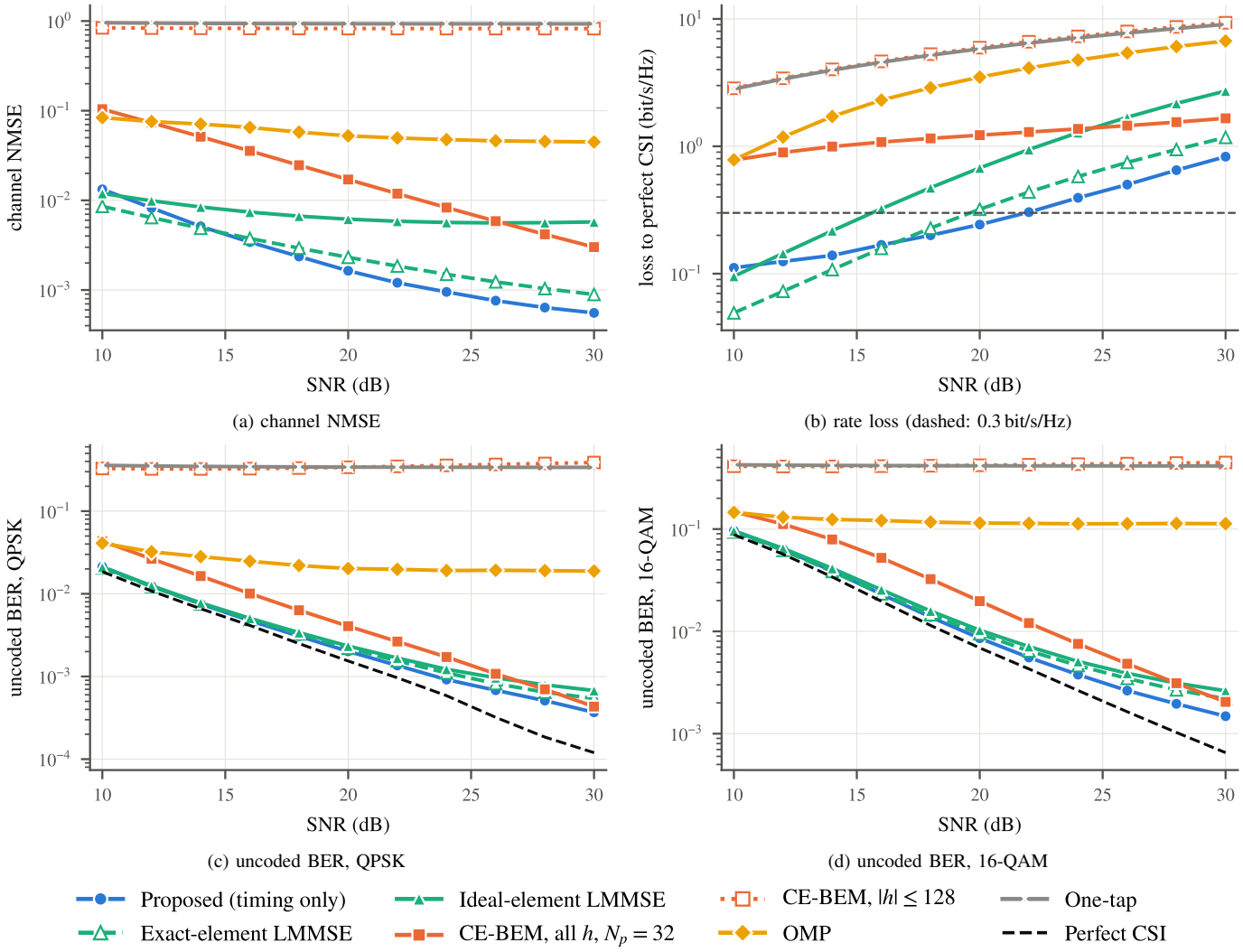
(d) uncoded BER, 16-QAM

Fig. 4. Estimators versus SNR in the format of the CE-BEM paper [6] (Figs. 10–11 there), shared-path channel, $N_p = 2$ (CE-BEM with all harmonics: $N_p = 32$), 30 new channels, SNR in 2 dB steps. Loss is to perfect CSI with the residual-based robust MMSE; BER after unbiased MMSE ($7.7 \cdot 10^5$ / $1.5 \cdot 10^6$ bits per point; points with no errors are omitted). At this bandwidth the flat-calibrated element LMMSE coincides with the exact-element curve to within 0.02 bit/s/Hz (Table II).

4) the 20/30 dB losses are 0.20/0.74 (ours), 0.26/1.02 (exact element), 0.63/2.55 (ideal element) and 1.21/1.64 (CE-BEM, $N_p = 32$). Figs. 4 and 5 show 10–30 dB in 2 dB steps on a separate set of channels. On the shared path ours has the lowest NMSE at every SNR (-23.7 to -41.3 dB) and loses at most 0.12 bit/s/Hz; the ideal-element LMMSE floors at -24 dB, and its BER with it. Our BER is within a factor of about two of perfect CSI for both modulations. On TDL-C ours crosses 0.3 bit/s/Hz at 22 dB as every estimator approaches a model-error floor. Beyond the plotted range (35–40 dB, same run) ours keeps the lowest NMSE and rate loss on both channels, but on TDL-C CE-BEM with eight times our pilots reaches the lower BER ($2.8 \cdot 10^{-4}$ against $7.1 \cdot 10^{-4}$ for 16-QAM at 40 dB).

Fig. 6 shows that the literature's estimator is not a weak baseline. As $\boldsymbol{H}$ is linear in Γ_s, the ideal and the physical element can be blended. With ideal elements ($\alpha = 0$) the ideal-element LMMSE is the exact model and matches ours (0.053 against 0.078 at 20 dB, within the intervals). As the non-ideality grows, its loss rises past 0.3 bit/s/Hz at $\alpha = 0.5$ (20 dB) and $\alpha = 0.25$ (30 dB), while ours stays at 0.07–0.08 (0.15–0.18 at 30 dB). An approximate element model is nearly as good as the exact one (the 5–20 MHz mismatch costs ≤ 0.4 dB of NMSE at 20 dB); assuming the ideal switch is what fails (Corollary 1). Over this 7.7 MHz band the element barely changes with frequency, so what α varies is the non-ideality ($\Gamma_1 \neq -\Gamma_2$, a static $\approx 7.7°$ phase error and 0.57 dB imbalance), not dispersion: two calibrated scalars per state remove the failure entirely (Table II). Fig. 6 therefore tests the ± 1 assumption; dispersion is tested at the full band (Sec. V-E).

Ours (Fig. 7) is the only estimator within 0.3 bit/s/Hz with at most 32 pilot symbols: two on the shared-path channel ($0.07\pm$



(a) channel NMSE

(b) rate loss (dashed: 0.3 bit/s/Hz)

(c) uncoded BER, QPSK

(d) uncoded BER, 16-QAM

Fig. 5. As Fig. 4 on the TDL-C channel (100 ns), $N_p = 4$.

0.02 / 0.11±0.03 at 20/30 dB), four on TDL-C. Its NMSE falls about 3 dB per doubling of N_p (-29.3 to -45.4 dB, $N_p = 1$–32, 20 dB); the ideal-element LMMSE saturates at -24.6 dB and CE-BEM needs $N_p \geq 8$. Paired per channel against the best of the ideal-element, CE-BEM and OMP baselines on that channel (minimum over those baselines and all equalizers; these do not include a calibrated element, see Table II), ours gains 0.41 ± 0.02 / 2.16 ± 0.04 bit/s/Hz at $N_p = 2$ on the shared-path channel and 0.30 ± 0.06 / 1.14 ± 0.25 on TDL-C; at every N_p from 2 to 32 the gain is 0.3–0.4 at 20 dB and 0.5–2.2 at 30 dB, with every interval above zero. A detector that estimates G_{mm} adds at most 0.15 to any loss. One pre-registered sanity check failed: truncated CE-BEM loses rate from 20 to 30 dB even with the oracle robust equalizer (its error sits on the missing harmonics).

Calibrated and structure-aware baselines. Table II (pre-registered) adds the baselines a skeptical reader would build. Two calibrated scalars per state close the whole gap between the ideal-element and the exact-element LMMSE over this band: the ideal-element estimator fails only through its ± 1 assumption. Ours matches the calibrated estimator at 20 dB (0.070 against 0.059 on the shared path, 0.280 against 0.323 on TDL-C) and is better at 30 dB, by 0.053 ± 0.035 (shared path) and 0.24 (TDL-C). The exact element with the same low rank is only 0.006–0.056 better than ours, so over a narrow band our advantage comes from the low rank, and not knowing the element costs little. OMP with the known kernels is competitive on the shared path; static pilots are good at low SNR there but floor, since the harmonic span is not flat, and fail on TDL-C. At 20 and 30 dB on both families ours is within 0.02 bit/s/Hz of the best of these practical estimators or better (pre-registered). The DPSS basis is not what makes it work: a plain delay grid with a flat prior does as well, and better on TDL-C. The rank is not critical (± 0.03). The delay window is: our hyperparameters are prior knowledge per channel family, and a window twice too long costs 0.14–1.10 bit/s/Hz, while half the window helps on the shared path and costs 0.48 on TDL-C at 30 dB. Choosing $\tau_{\max}$ by evidence, as Algorithm 2

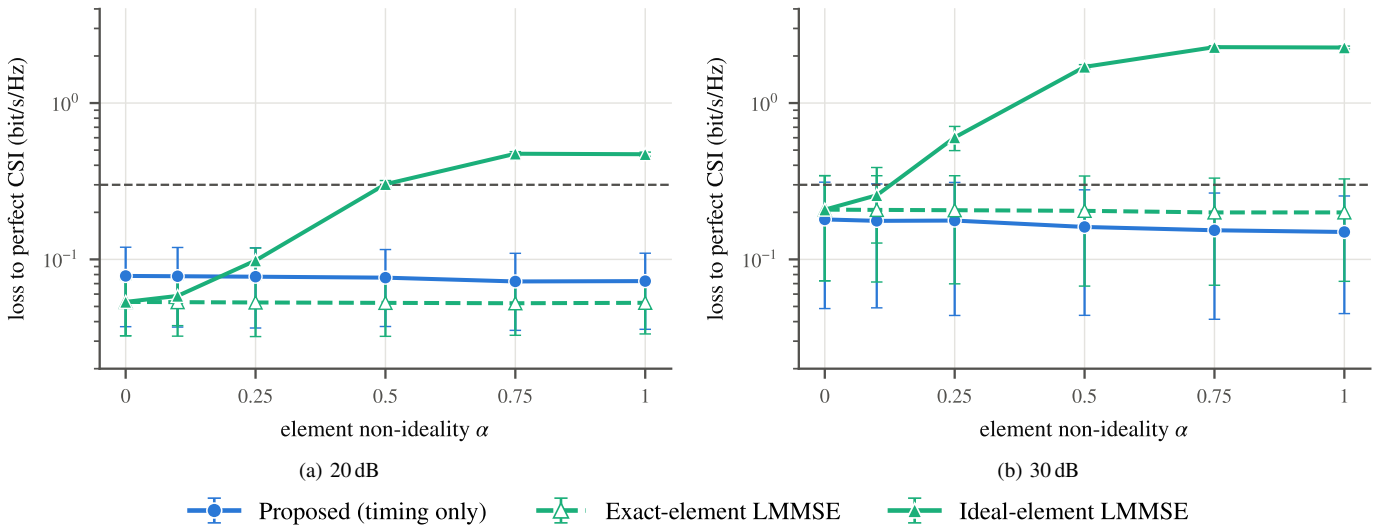


Fig. 6. Loss versus element non-ideality, $\boldsymbol{H}_\alpha = (1-\alpha)\boldsymbol{H}_{\text{ideal}} + \alpha \boldsymbol{H}_{\text{OpenRIS}}$ (shared path, $N_p = 2$, 30 channels, 95% intervals). CE-BEM ($|h| \le 128$) and one-tap lose more than 6 bit/s/Hz at every α.

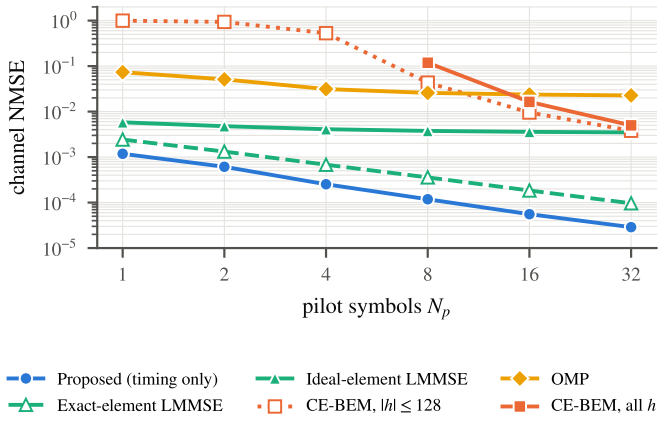


Fig. 7. NMSE versus pilot symbols, shared-path channel, 20 dB, 30 channels. CE-BEM with all harmonics is run from $N_p = 8$.

chooses the window start, is a natural extension that we have not tested.

B. Learning-based baseline and a self-trained prior

Table III answers whether learning would do better. It would, if trained on the right channels: the learned-prior LMMSE beats ours on every family when trained on the same distribution, already with $N_{\text{tr}} = 100$ (0.006 against 0.070 at 20 dB on the shared path with $N_{\text{tr}} = 1000$). The reason is that our simulated families are low-dimensional. At $M = 256$ the band is 7.7 MHz, so a 30 ns channel with a strong line of sight is nearly flat (the 1000 shared-path channels span 6 principal components, with mean correlation 0.92), and TDL has fixed tap delays (50 components). A prior learned on one family transfers poorly to a different delay spread or Rician factor: trained on the 30 ns family and tested at 100 ns it is worse than ours by 0.22 ± 0.17 / 0.95 ± 0.46 bit/s/Hz, and trained on TDL-C it ties ours on TDL-D at 20 dB and is worse by 0.37 ± 0.20 at 30 dB; between the similar TDL-A and TDL-C it still gains 0.06 ± 0.03 at 20 dB. Our estimator needs neither training channels nor an element model, and it can supply the training data itself: a prior learned from its 8-pilot estimates of 300 channels beats it by 0.042 ± 0.012 (shared path) and 0.107 ± 0.027 bit/s/Hz (TDL-C) at 20 dB. Because such a prior is collected in place, it matches the deployment distribution by construction; we offer it as an optional second stage.

TABLE II
CALIBRATED AND STRUCTURE-AWARE BASELINES, ABLATIONS AND SENSITIVITY: LOSS TO PERFECT CSI (BIT/S/HZ) AT 20/30 DB, 30 NEW CHANNELS PER FAMILY ($N_p = 2$ SHARED PATH, 4 TDL-C). "OURS" IS ALGORITHM 1.

Estimator	Shared path	TDL-C
Ours	.070/.146	.280/.977
Exact element, low rank (reference)	.063/.126	.258/.921
Exact element, linear (reference)	.060/.203	.325/1.23
Flat-calibrated element	.059/.199	.323/1.22
Flat element, ratio estimated	.067/.255	.359/1.48
Ideal (± 1) element	.478/2.28	.680/2.69
OMP with known kernels	.061/.187	.317/1.38
Per-state static pilots	.074/.521	1.44/4.07
CE-BEM, all h, $N_p = 32$	.422/.552	1.20/1.68
Ours, linear stage only	.094/.250	.355/1.28
Ours, delay-grid basis (no DPSS)	.077/.110	.241/.768
Ours, $\tau_{\max} \times 0.5$	.033/.065	.306/1.46
Ours, $\tau_{\max} \times 2$	.213/.356	.663/2.08
Ours, $r = 4$ (shared) / 8 (TDL)	.065/.130	.253/.985
Ours, $r = 16$ (shared) / 24 (TDL)	.094/.177	.289/.999

C. No element model: robustness beyond the generating model

A receiver tested on data from its own model can look better than it is. Table IV runs the estimator *unchanged* on data that violate its structure, paired with the same channel without the change. In every case the extra loss stays within 0.05 bit/s/Hz,

TABLE III
LOSS TO PERFECT CSI (BIT/S/HZ) AT 20/30 DB AGAINST THE LEARNED-PRIOR LMMSE ($N_{\mathrm{tr}} = 300$, 30 TEST CHANNELS; $N_p = 2$ SHARED PATH, 4 TDL). SHIFTED: TRAINED ON THE FAMILY IN BRACKETS. SELF-TRAINED: PRIOR LEARNED FROM OUR OWN ESTIMATES OF 300 CHANNELS WITH 8 PILOT SYMBOLS AT 20 DB.

Test channels	Ours	Learned	Learned, shifted	Self-trained
Shared, 30 ns, K 10 dB	.071/.143	.007/.031	–	.029/.118
Shared, 100 ns, K 3 dB	.124/.468	.054/.247	.346/1.42 (30 ns)	–
TDL-C	.185/.668	.058/.213	–	.078/.362
TDL-A	.178/.726	.047/.184	.121/.611 (C)	–
TDL-D	.483/1.68	.043/.204	.445/2.05 (C)	–

TABLE IV
ESTIMATOR UNCHANGED ON DATA VIOLATING ITS MODEL: RELATIVE CHANGE OF $\boldsymbol{H}$ AND EXTRA LOSS (BIT/S/HZ, PAIRED, 30 CHANNELS).

Violation	$\Delta\boldsymbol{H}$	20 dB	30 dB
Time-domain truth	0.006	-0.004 ± 0.005	$+0.006 \pm 0.009$
Switching transients	0.001	$+0.001 \pm 0.001$	$+0.003 \pm 0.006$
State-dep. output, $\epsilon = 0.2$	0.40	≤ 0.004	$+0.034$
Joint-state coupl., $\kappa = 0.08$	0.29	≤ 0.002	$+0.015$
Resonance spread 10 MHz	0.18	$+0.003 \pm 0.008$	$+0.046 \pm 0.044$
Beam squint, $Q = 256$	0.016	≤ 0.001	-0.002

which passes the pre-registered bar of 0.3. Element-to-element resonance spread of 5–20 MHz costs 0.06–0.15 bit/s/Hz. A six-parameter fit of the element model is not better than timing-only ($+0.08 \pm 0.05$ worse at 30 dB) and fails with element spread ($+0.73 \pm 0.43$). On a 16-element surface the linear first stage provides most of the gain (loss 0.10/0.23 at $N_p = 2$); the low-rank step and one decision-directed pass reduce it to 0.07/0.11 and 0.02/0.03. The low-rank property itself matters most for large surfaces: the effective rank of the true element-domain matrix (99% energy, CDL-C, 98 MHz, beam squint) is 5, 9, 13 and 16 for $Q = 16$, 64, 256 and 1024.

D. Timing

Table V tests Algorithm 2 with physical bulk delays ($\tau_b \sim \mathcal{U}(50, 300)$ ns, $\tau_u \sim \mathcal{U}(30, 300)$ ns). A BS-clock-locked RIS needs the search, and with the search the rate is within noise of known timing. An incident-signal-locked RIS needs no search, and running it anyway costs nothing significant. A synchronization error of up to ± 150 ns breaks a fixed delay window, but not the jointly chosen one. Is the evidence necessary? On a fine grid the pilot residual of the full low-rank estimate also finds the offset with two or more pilot symbols (not significantly worse). With a single pilot symbol it overfits: it is worse by 1.49 ± 0.56 / 2.83 ± 1.19 bit/s/Hz, while the evidence stays within 0.03 of known timing at 20 dB. The evidence is also 5–58 times cheaper per search (3.2 vs 17.2 s at $M = 256$; 11.7 vs 674 s on TDL-C), because it needs no low-rank estimate per candidate. Random per-edge jitter cannot be estimated. It must stay below about 4 ns rms at 30 dB (20 ns at 20 dB), whereas ordinary clocks have picosecond jitter.

TABLE V
TIMING: LOSSES AND PAIRED DIFFERENCES (BIT/S/HZ, 30 CHANNELS).

	20 dB	30 dB
BS-clock RIS, offset ignored	1.62	4.23
Alg. 2 minus known offset	$+0.011 \pm 0.019$	$+0.033 \pm 0.066$
Incident-locked: Alg. 2 minus none	$+0.001 \pm 0.007$	$+0.040 \pm 0.082$
Sync error ≤ 150 ns, fixed window	0.22	1.16
window chosen by evidence	0.05 ± 0.05	0.05 ± 0.08
$N_p = 1$: residual minus evidence	1.49 ± 0.56	2.83 ± 1.19

E. Full band and coded link

At the full band ($M = 3276$, 30 channels) the SER with two pilot symbols is 1.10 [1.06, 1.15] times the perfect-CSI SER at 20 dB with 16-QAM and 1.37 [1.18, 1.86] times at 28 dB with 64-QAM (95% clustered-bootstrap intervals). One decision-directed pass brings these to 1.05 and 1.12. CE-BEM ($|h| \leq 8$) is 18 and 32 times worse; the one-tap receiver's SER is 0.84 and 0.96.

Element-model baselines where the element is dispersive. Across 98 MHz each state's phase drifts by about 47° and Γ_2/Γ_1 varies by up to 11.5° and ±7%. On 30 new channels (pre-registered), matrix-free, the flat-calibrated, flat-estimated, exact-element (linear) and ideal-element LMMSEs have 1.49 [1.21, 2.27], 1.59 [1.26, 2.46], 1.67 [1.31, 2.65] and 2.10 [1.59, 3.46] times our SER at 28 dB with 64-QAM and two pilot symbols (paired bootstrap), and 1.09–1.19 times at 20 dB with 16-QAM. One pilot symbol suffices: our SER is 1.10, 1.06 and 1.03 times the perfect-CSI SER with one, two and four pilot symbols at 20 dB. The mechanism is again mainly the low rank. The exact-element LMMSE is no better than the flat-calibrated one, because with our 0.2 μs delay prior every linear estimator floors near -34 dB NMSE even at 60 dB SNR (the channel's energy sits in the DPSS components at the window edge, where the prior variance falls off). Dispersion alone limits a calibrated flat model to -36 to -40 dB, below that floor, so it binds only at high SNR or with a better-matched prior; ours reaches -37.5 dB at 28 dB with four pilot symbols. For the coded link (a rate-1/2 convolutional code, 16-QAM, max-log LLRs from each receiver's own estimate, and 2000 blocks per point) the proposed receiver (Fig. 8) crosses 10% block error rate (BLER) at 8.51 dB against 8.37 dB with perfect CSI, a gap of 0.14 dB [0.09, 0.20]. One-tap never decodes, and truncated CE-BEM floors at 34%.

F. Mobility and pilot overhead

When the user moves, only the RIS–user hop changes. Each path rotates by $e^{j2\pi\nu_\ell n T_s}$ between symbols and also mixes subcarriers within a symbol. That mixing is -60, -47 and -34 dB below the signal at 3, 30 and 120 km/h, so the main effect is the ageing of the estimate. With N_p pilot symbols every N symbols, the net rate is $R_{\mathrm{net}} = \frac{1}{N}\sum_{n=N_p}^{N-1} R(n)$, where $R(n)$ uses the estimate on symbol n with one common phase corrected per symbol, as 5G phase-tracking pilots do. Table VI selects (N_p, N) and, for CE-BEM, the harmonic range on 20 channels, and reports them on 20 *other* chan-

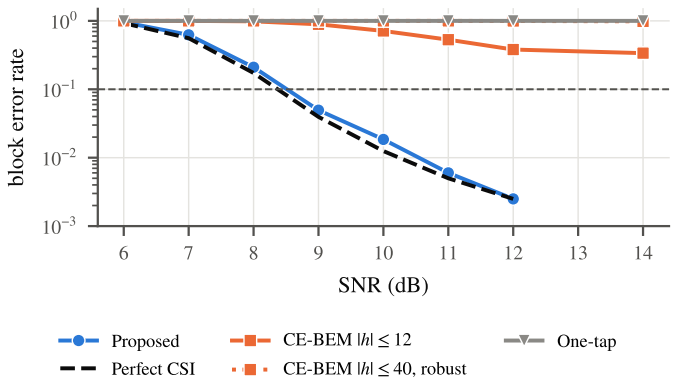


Fig. 8. Coded BLER, $N_p = 2$, 2000 blocks per point. Gap to perfect CSI at 10% BLER: 0.14 dB [0.09, 0.20].

TABLE VI
NET THROUGHPUT AFTER PILOT OVERHEAD (BIT/S/HZ), (N_p, N) SELECTED ON 20 CHANNELS AND REPORTED ON 20 DISJOINT CHANNELS; OURS − CE-BEM PAIRED, 95% INTERVAL.

		20 dB			30 dB		
Channel	km/h	ours	BEM	diff.	ours	BEM	diff.
Shared	3	6.24	5.24	1.01 ± 0.14	8.99	7.34	1.65 ± 0.34
Shared	30	5.97	4.40	1.57 ± 0.30	8.56	5.42	3.14 ± 0.40
Shared	120	5.26	3.15	2.11 ± 0.30	7.30	3.38	3.92 ± 0.53
TDL-A	3	5.38	3.71	1.67 ± 0.17	7.37	4.88	2.49 ± 0.21
TDL-A	30	4.56	1.60	2.97 ± 0.13	5.51	1.68	3.84 ± 0.27
TDL-A	120	3.81	1.08	2.73 ± 0.28	4.35	1.07	3.27 ± 0.58
TDL-C	3	5.34	3.61	1.74 ± 0.16	7.23	4.81	2.41 ± 0.27
TDL-C	30	4.53	1.65	2.88 ± 0.29	5.53	1.81	3.72 ± 0.42
TDL-C	120	3.88	0.75	3.13 ± 0.31	4.61	0.74	3.88 ± 0.54
TDL-D	3	5.17	3.61	1.56 ± 0.17	6.80	4.75	2.05 ± 0.41
TDL-D	30	4.65	2.17	2.48 ± 0.25	5.85	2.39	3.46 ± 0.42
TDL-D	120	3.92	1.15	2.77 ± 0.31	4.68	1.15	3.53 ± 0.69

nels. Ours beats the best CE-BEM in all 24 cases, by 1.0–3.9 bit/s/Hz, every interval above zero (CE-BEM selects 16–32 pilot symbols). Selecting on the reported channels changes the numbers by at most 0.13. Net of pilots, ours keeps 75–93% of the perfect-CSI rate on the shared path and 73–88% on TDL at 3 km/h, but only 46–66% on TDL at 120 km/h: with a 30-symbol coherence time the selected design spends one pilot symbol in every 6. Slow channels select two (shared) or four (TDL) pilot symbols per 40–120 symbols. High mobility is thus a limit of pilot-based estimation, not of the coupling model. Two decision-directed trackers (LMMSE and Kalman) were worse than reusing the pilot estimate, because wrong decisions accumulate after about ten symbols.

G. Hardware, deployment and the cost of time modulation

Impairments. Residual CFO must stay below about 1% of the spacing (0.10/0.33 bit/s/Hz at 20/30 dB), and phase noise below about 100 Hz linewidth or be tracked (0.16/0.33 at 100 Hz). Rise times up to 42 ns need no calibration (≤ 0.011 at 30 dB); longer ones are calibrated or fitted blindly from the pilots (within 0.03 slot). Rise time also sets the emission: ideal edges give an adjacent-channel leakage ratio of 36.8 dB, below the 45 dB required of a base station [16], and 0.05-slot edges give 52.7 dB. Slow edges shrink our lead: at a 0.3-slot ramp CE-BEM with eight pilot symbols reaches 8.2 against our 8.56 bit/s/Hz with two.

Deployment. A direct path needs no change ($T_{q,1} + T_{q,2} = \boldsymbol{I}$; 5.28/8.13 against 5.39/8.49 bit/s/Hz with perfect CSI), nor do two panels with different patterns (5.54/8.67 against 5.64/8.86). An unknown OFDM interferer at 0–20 dB SIR costs 0.03–0.17; 3GPP CDL channels match TDL within 0.07.

Cost of time modulation. At equal transmit power and 20 dB, a TM-RIS with a pattern designed for the user gives 0.68/1.98/3.15 bit/s/Hz with one-tap for $D = 0.625/0.75/0.875$, 2.98/4.06/5.35 with ours, and 3.08/4.18/5.50 with perfect CSI; a static RIS optimized for the user gives 6.66. Time modulation is thus not a single-user rate gain but is used for security, harmonic steering or sensing, and our receiver recovers its cost to within 0.1–0.2 bit/s/Hz of perfect CSI. An eavesdropper with generic CE-BEM still obtains 1.8–2.2 bit/s/Hz (2.5–3.0 knowing the pattern), so the pattern must be kept secret.

VI. CONCLUSION

Grouping the harmonic expansion of a dispersive TM-RIS by element state separates the coupling exactly into a part the receiver knows, the switching timing, and a part it learns, the propagation together with the element dispersion. A receiver built on this separation needs no element model. It estimates the coupling from two to eight pilot symbols, finds the RIS clock offset by itself, and equalizes without forming the coupling matrix. In pre-registered tests at 20–30 dB it is within 0.02 bit/s/Hz of, or better than, every estimator that needs no training data, including element-model estimators with a calibrated element. Over a narrow band two calibrated scalars per state make an element model as good as the exact one, and our advantage there comes from the low rank; over the full band the calibrated estimators have 1.5 times our symbol error rate at 28 dB, again mainly because of the low rank. A prior learned from channels of the deployment distribution is better, a prior learned elsewhere can be much worse, and a prior learned from our own estimates keeps most of the gain without ground truth. It decodes within 0.14 dB of perfect CSI. Its limits are stated above: the receiver must know every panel's pattern (A1); delay spreads whose cascade exceeds the CP are out of scope (A2); at 120 km/h on TDL channels pilot overhead and estimate ageing leave 46–66% of the perfect-CSI rate; below about 17–22 dB a receiver given the element model is marginally better (by ≤ 0.06 bit/s/Hz); the delay window $\tau_{\max}$ must suit the channel family (twice too long costs up to 1.1 bit/s/Hz); the stated requirements on jitter, CFO, phase noise and emission apply; and the validation is by simulation with one published cell and no mutual coupling. Over-the-air validation with a real time-modulated surface is the next step.

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